

COORDINATION CHEMISTRY: MASTER TEXTBOOK

1. Introduction

Coordination Compounds

A coordination compound contains a central metal atom or ion surrounded by a number of oppositely charged ions or neutral molecules (ligands).

Key Terms: Coordination Entity, Central Atom, Ligand, Coordination Number, Coordination Sphere.

2. Werner's Coordination Theory

Primary and Secondary Valency

Werner proposed that metals exhibit two types of valencies:

- **Primary Valency:** Corresponds to the oxidation state of the metal; ionizable.
- **Secondary Valency:** Corresponds to the coordination number; non-ionizable.

3. Bonding Theories

Valence Bond Theory (VBT)

VBT explains the formation of coordinate bonds through orbital hybridization (e.g., d^2sp^3 for octahedral). It helps predict the magnetic nature (paramagnetic/diamagnetic) and geometry.

Crystal Field Theory (CFT)

CFT treats the bond between metal and ligand as purely electrostatic. It explains the splitting of d-orbitals into energy levels (e.g., t^2_g and e_g in octahedral complexes).

4. Isomerism

Structural and Stereoisomerism

Structural: Ionization, Hydrate, Linkage, Coordination isomerism.

Stereo: Geometrical (cis-trans) and Optical isomers.